

Section 3.2 Determinants and Matrix Inverses

Exercise 3.2.2

Use determinants to find which real values of c make each of the following matrices invertible.

$$A = \text{a. } \begin{bmatrix} 1 & 0 & 3 \\ 3 & -4 & c \\ 2 & 5 & 8 \end{bmatrix} \Rightarrow 1(-32 - 5c) - 0 + 3(15 + 8) = -32 - 5c + 45 + 24$$

$$= 37 - 5c$$

$$5c = 37$$

$$c = \frac{37}{5}$$

$\therefore$ matrix A is invertible
for $c \neq \frac{37}{5}$

$$\text{b. } \begin{bmatrix} 0 & c & -c \\ -1 & 2 & 1 \\ c & -c & c \end{bmatrix} \quad 0(\dots) - c(-c - c) + (-c)(c - 2c)$$

$$c^2 + c^2 \Rightarrow 2c^2 = 0$$

$$c \neq 0$$

Test Test Test

c. $\begin{bmatrix} c & 1 & 0 \\ 0 & 2 & c \\ -1 & c & 5 \end{bmatrix} \Rightarrow c(10 - c^2) - 1(0 + c) + 0$
 $+ 10c - c^3 - c = 0$
 $\Rightarrow 9c - c^3 = 0$

$c(9 - c^2) = 0 \therefore c = \{\pm 3, 0\}$
 $c^2 = 9$
 $c = \pm 3$

$\therefore$! invertible
 for $c \neq \pm 3, 0$,
 invertible for
 $c \in \mathbb{N}, c \neq \{\pm 3, 0\}$

d. $\begin{bmatrix} 4 & c & 3 \\ c & 2 & c \\ 5 & c & 4 \end{bmatrix} \Rightarrow 4(8c - c^2) - c(4c - 5c) + 3(c^2 + 10)$
 ~~$32c - 4c^2 - 16c^2 + 5c^2 + 3c^2 + 30$~~
 ~~$-20c^2 - 15c - 12c^2$~~
 ~~$-12c^2 + 12c + 30 = 0$~~
 ~~$6 \quad 16 \quad 15$~~
 ~~4~~

32

$4(8c^2) - c(-c) + 3(c^2 - 10)$
 $\downarrow$
 $32 - 4c^2 + c^2 + 3c^2 - 30$
 $= 2 + 0$
 $= 2$

$\therefore A$ is invertible
 for $c \neq 2$

Exercise 3.2.3

Let A , B , and C denote $n \times n$ matrices and assume that $\det A = -1$, $\det B = 2$, and $\det C = 3$. Evaluate:

a. $\det(A^3 B C^T B^{-1})$

$$-1^3 (2) (3) \left(\frac{1}{2}\right)$$

$$6 \rightarrow 3 \rightarrow \boxed{-3}$$

b. $\det(B^2 C^{-1} A B^{-1} C^T)$

$$\boxed{-2}$$

$$(4) \left(\frac{1}{3}\right) (-1) \left(\frac{1}{2}\right) (3)$$

$$\begin{array}{c} \swarrow \quad \searrow \\ 2 \quad 2 \\ \swarrow \quad \searrow \\ 4 \quad 2 \\ \swarrow \quad \searrow \\ \boxed{-2} \end{array}$$

Exercise 3.2.5

If A is 3×3 and $\det(2A^{-1}) = -4$ and $\det(A^3(B^{-1})^T) = -4$, find $\det A$ and $\det B$.

$$\det \cancel{A^{-1}} = -2 \quad + 2^3 \det(A^{-1}) = -4$$

$$\det \cancel{A} = 2$$

$$A = -\frac{1}{2} \quad \frac{2^3}{\det A} = -4, \frac{1}{8}A = -4, A = \boxed{-2}$$

$$\det(8(\tilde{B}^{-1})^T) = -4$$

$$\det(\tilde{B}^{-1})^T = -\frac{1}{4}$$

$$\boxed{B = \frac{1}{2} \cdot 2}$$

$$\left. \begin{array}{l} \therefore \det A = -2 \\ \det B = 2 \end{array} \right\} \checkmark$$

Exercise 3.2.6

Let $A = \begin{bmatrix} a & b & c \\ p & q & r \\ u & v & w \end{bmatrix}$ and assume that $\det A = 3$. Compute:

a. $\det(2B^{-1})$ where $B = \begin{bmatrix} 4u & 2a & -p \\ 4v & 2b & -q \\ 4w & 2c & -r \end{bmatrix}$

$\det(C)$

multi $\rightarrow 4(2(-1)) = -8$, move 2 out of det

$\Rightarrow \det(B^{-1}) = 8 \leftarrow \text{from } 2^3, ka \Rightarrow k^n$

$\det(B) = -\frac{1}{8}$

1) pull
2) n^{-1} afterwards,
so B is some mod of A

Todo!!

Finding 'B'

1) From $4(2(-1)) = -8$

2) Transpose stuff

$\Rightarrow -8(3) = -24$

$2^3 = 8 \frac{1}{3} = -24$

$\Rightarrow B = \frac{-24}{-8} = 3$

$B = 24 \quad B = \frac{-8}{24} = -\frac{1}{3}$

$\Rightarrow \frac{-8}{24} = -\frac{1}{3} \Rightarrow -\frac{1}{3}$

o) Some Row addition?

1) Transpose

2) $2(-1(3))$

3) $2 \cdot 1^{-1}$ terms

4) Swap

$\therefore \det(2C^{-1}) = \frac{4}{9}$

1) $C = 3$, odd T do nothing

2) $C = 3(3(2))(-1) = -18$

3) 18 = row swap

4) $C = 18$

matrix size

Then $2 \cdot 1^{-1} \Rightarrow 2^3 C = \frac{1}{8}$, or $3 \cdot 18$

$2^3 \det(C^{-1}) = 18$

$C^{-1} = \frac{8}{18}$

$$\Rightarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

e. $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 6 & -6 \\ 1 & 2 & -1 \end{bmatrix}$

1) $\det(\lambda I - A)$

$\Rightarrow \begin{bmatrix} \lambda-1 & 2 & -3 \\ -2 & \lambda-6 & 6 \\ -1 & -2 & \lambda+1 \end{bmatrix}$

2) Find $\det \Rightarrow$

$$\begin{bmatrix} \lambda-1((\lambda-6)(\lambda+1) - (-12)) \\ 2(-2(\lambda+1) + 6) \\ -3[4 + (\lambda-6)] \end{bmatrix} = \begin{bmatrix} \lambda-1(\lambda^2-5\lambda+6) \\ 4\lambda-8 \\ -3\lambda+6 \end{bmatrix} = \begin{bmatrix} (\lambda-1)(\lambda-2)(\lambda-3) \\ 4(\lambda-2) \\ -3(\lambda-2) \end{bmatrix}$$

NOT diagonalizable;
req 3 linear ind
eigenvalues!!
only have 2!

Then combine w/

3) combining: $[(\lambda-1)(\lambda-2)(\lambda-3) + 4(\lambda-2) - 3(\lambda-2)] = p$

Pulling out $(\lambda-2)$ set; $(\lambda-2)[\lambda^2-4\lambda+4]$, so char polynomial
With eigen value $\lambda=2$
 $p = (\lambda-2)^3$

Eigen vector, solve $(\lambda I - A)x = 0$. Then $\begin{bmatrix} \lambda-1 & 2 & -3 \\ -2 & \lambda-6 & 6 \\ -1 & -2 & \lambda+1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -3 \\ -2 & -4 & 6 \\ -1 & -2 & 3 \end{bmatrix}$
s.t $v_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$ $v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ For obtain v_3
 $\Rightarrow x_1 + 2x_2 - 3x_3 = 0$